Phishing URL Detection — Hybrid ML + Deep Learning

This notebook demonstrates:

- Random Forest baseline model
- BiLSTM deep learning model
- Fusion ensemble model
- Cross-dataset benchmarking
- Reproducible evaluation pipeline

Dataset: StealthPhisher + Kaggle Balanced URLs

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import os
print(os.listdir('/content/drive/MyDrive'))
```

```
['Colab Notebooks', 'songs', 'IMG20230709173734.jpg', 'IMG20230709173708.jpg', 'IMG20230709173706.jpg', 'IMG20230709173441.jpg
```

```
import os
os.listdir('/content/drive/MyDrive')
```

Show hidden output

```
!unzip /content/drive/MyDrive/PhishingProject.zip -d /content
```

Show hidden output

```
!unzip /content/drive/MyDrive/PhishingProject.zip -d /content
```

Show hidden output

```
%cd /content/PhishingProject
!ls
```

```
/content/PhishingProject
benchmark_split.py       cross_eval.py     rf_preds.npy
benchmark_test.csv       datasets          rf_true.npy
benchmark_test_idx.npy   evaluate.py       save_bilstm_preds.py
benchmark_train.csv      full_plots.py     show_results.py
benchmark_train_idx.npy  fusion_eval.py    train.py
bilstm_benchmark.py      loss_history.npy  vocab.json
bilstm_model.pt          plots.py          X_test.npy
bilstm_preds.npy         results.json      y_test.npy
bilstm_true.npy          rf_benchmark.py
```

🔐 Phishing URL Detection Benchmark

Dataset

- StealthPhisher Dataset
- Kaggle Balanced URLs Dataset

Models Compared

- Random Forest (Feature-based ML)
- BiLSTM Neural Network (Deep Learning)
- Fusion Model (Ensemble)

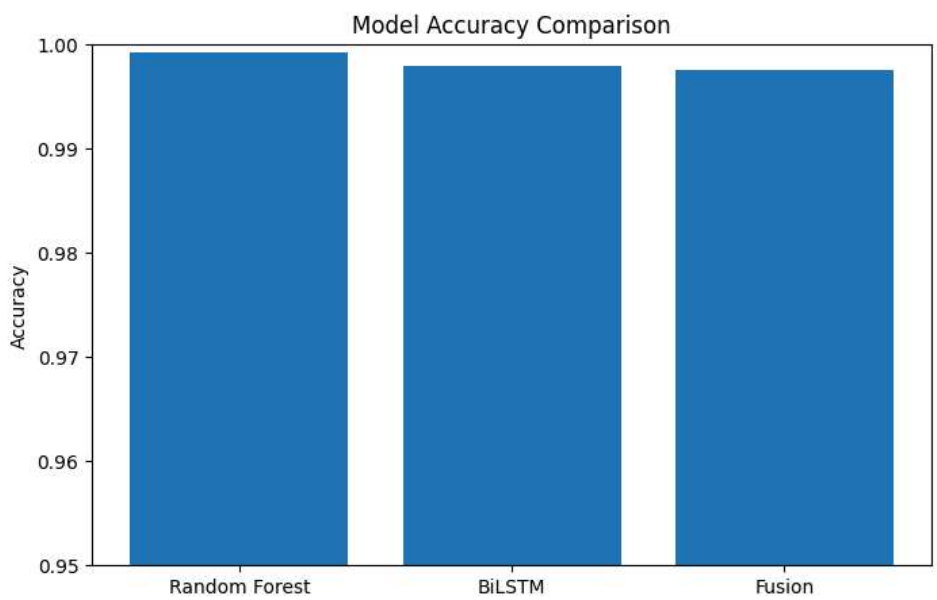
Experiment Pipeline

1. Load benchmark split
2. Evaluate Random Forest
3. Evaluate BiLSTM
4. Fusion evaluation
5. Visual comparison

```
import numpy as np
from sklearn.metrics import accuracy_score

# Load saved predictions
rf_preds = np.load("rf_preds.npy")
rf_true = np.load("rf_true.npy")

bilstm_preds = np.load("bilstm_preds.npy")
bilstm_true = np.load("bilstm_true.npy")

rf_acc = accuracy_score(rf_true, rf_preds)
bilstm_acc = accuracy_score(bilstm_true, bilstm_preds)

fusion_preds = np.round((rf_preds + bilstm_preds) / 2)
fusion_acc = accuracy_score(rf_true, fusion_preds)

print("=== Model Accuracies ===")
print(f"Random Forest: {rf_acc:.6f}")
print(f"BiLSTM:        {bilstm_acc:.6f}")
print(f"Fusion:        {fusion_acc:.6f}")
```

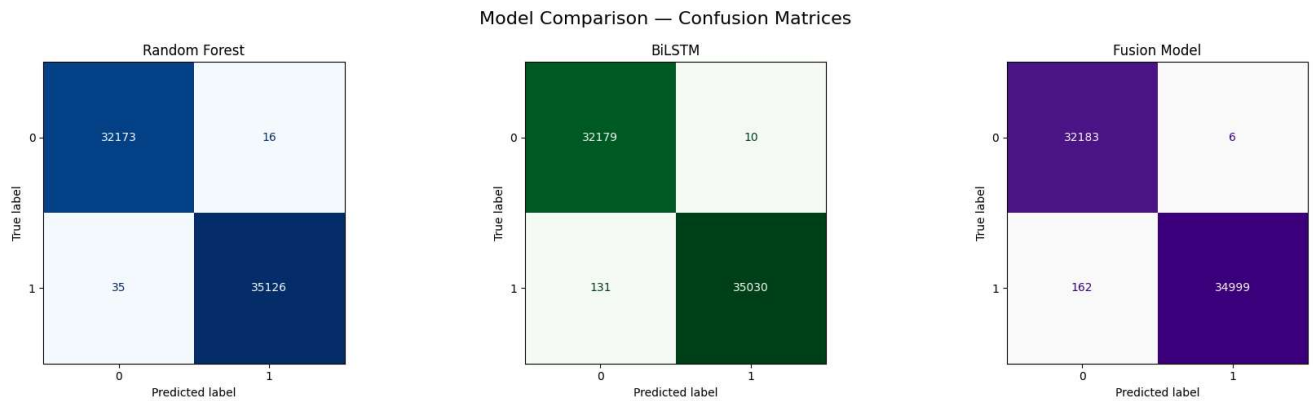
```
=== Model Accuracies ===
Random Forest: 0.999243
BiLSTM:        0.997906
Fusion:        0.997506
```

```
import matplotlib.pyplot as plt

models = ["Random Forest", "BiLSTM", "Fusion"]
accuracies = [rf_acc, bilstm_acc, fusion_acc]

plt.figure(figsize=(8,5))
plt.bar(models, accuracies)
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.ylim(0.95, 1.0)
plt.show()
```




```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

# Compute confusion matrices
```

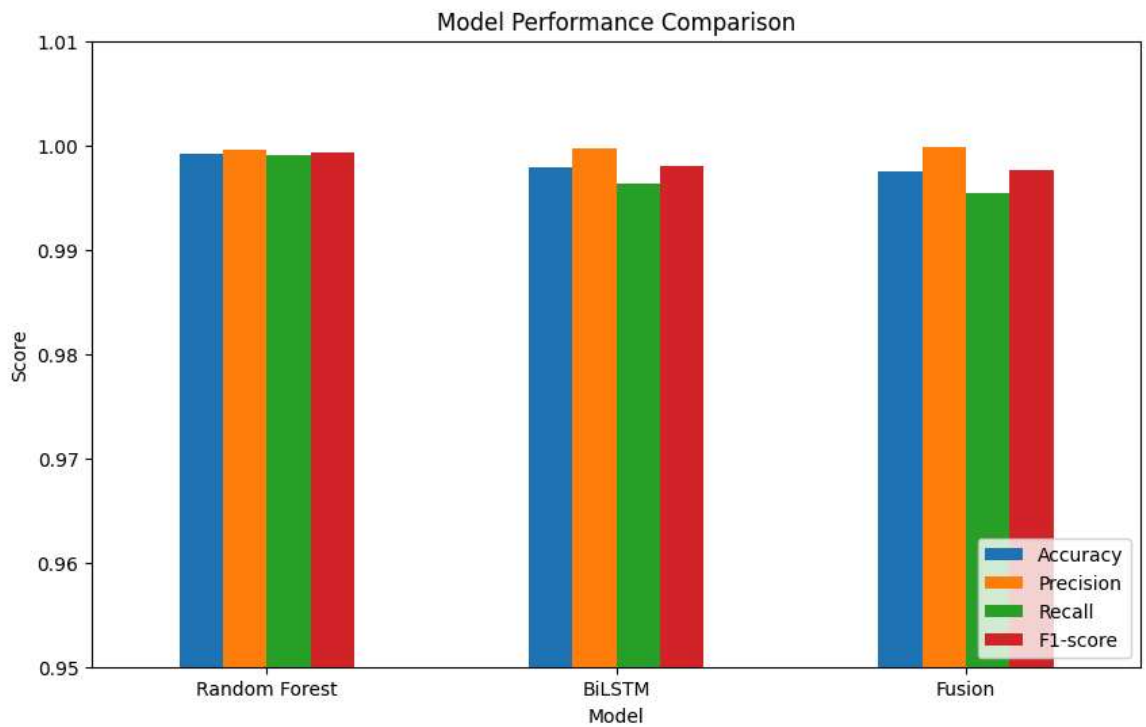
```
rf_cm = confusion_matrix(rf_true, rf_preds)
bilstm_cm = confusion_matrix(bilstm_true, bilstm_preds)
fusion_cm = confusion_matrix(rf_true, fusion_preds)

# Create subplot figure
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# RF
ConfusionMatrixDisplay(rf_cm).plot(ax=axes[0], cmap="Blues", colorbar=False)
axes[0].set_title("Random Forest")

# BiLSTM
ConfusionMatrixDisplay(bilstm_cm).plot(ax=axes[1], cmap="Greens", colorbar=False)
axes[1].set_title("BiLSTM")

# Fusion
ConfusionMatrixDisplay(fusion_cm).plot(ax=axes[2], cmap="Purples", colorbar=False)
axes[2].set_title("Fusion Model")

plt.suptitle("Model Comparison — Confusion Matrices", fontsize=16)
plt.tight_layout()
plt.show()
```




```
from sklearn.metrics import precision_score, recall_score, f1_score
import pandas as pd

# Compute metrics
metrics = {
    "Model": ["Random Forest", "BiLSTM", "Fusion"],
    "Accuracy": [
        accuracy_score(rf_true, rf_preds),
        accuracy_score(bilstm_true, bilstm_preds),
        accuracy_score(rf_true, fusion_preds),
    ],
    "Precision": [
        precision_score(rf_true, rf_preds),
        precision_score(bilstm_true, bilstm_preds),
        precision_score(rf_true, fusion_preds),
    ],
    "Recall": [
        recall_score(rf_true, rf_preds),
        recall_score(bilstm_true, bilstm_preds),
        recall_score(rf_true, fusion_preds),
    ],
    "F1-score": [
        f1_score(rf_true, rf_preds),
        f1_score(bilstm_true, bilstm_preds),
        f1_score(rf_true, fusion_preds),
    ],
}

df_metrics = pd.DataFrame(metrics)
df_metrics
```

	Model	Accuracy	Precision	Recall	F1-score
0	Random Forest	0.999243	0.999545	0.999005	0.999275
1	BiLSTM	0.997906	0.999715	0.996274	0.997991

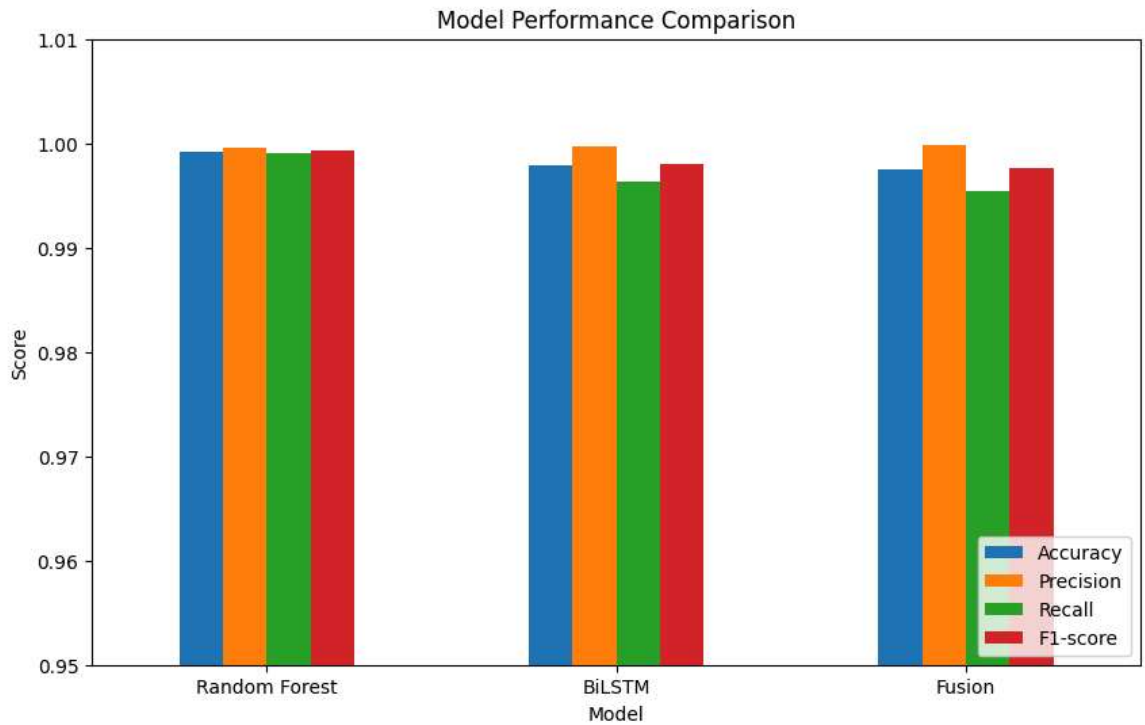
```
import matplotlib.pyplot as plt

# Set model names as index
df_plot = df_metrics.set_index("Model")

# Plot grouped bar chart
df_plot.plot(kind="bar", figsize=(10, 6))

plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.ylim(0.95, 1.01)
plt.xticks(rotation=0)
plt.legend(loc="lower right")

plt.show()
```




```
# Save metrics table
df_metrics.to_csv("performance_metrics.csv", index=False)

print("Metrics table saved as performance_metrics.csv")
```

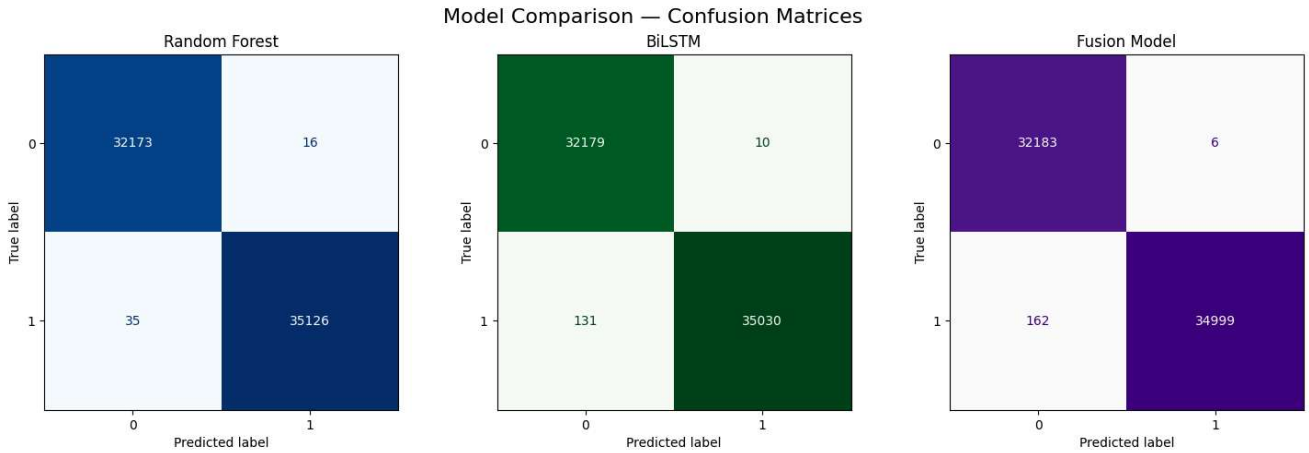
```
Metrics table saved as performance_metrics.csv
```

```
df_plot.plot(kind="bar", figsize=(10, 6))

plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.ylim(0.95, 1.01)
plt.xticks(rotation=0)
plt.legend(loc="lower right")

plt.savefig("performance_chart.png", dpi=300, bbox_inches="tight")
plt.show()

print("Performance chart saved as performance_chart.png")
```




```
Performance chart saved as performance_chart.png
```

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

ConfusionMatrixDisplay(rf_cm).plot(ax=axes[0], cmap="Blues", colorbar=False)
axes[0].set_title("Random Forest")

ConfusionMatrixDisplay(bilstm_cm).plot(ax=axes[1], cmap="Greens", colorbar=False)
axes[1].set_title("BiLSTM")

ConfusionMatrixDisplay(fusion_cm).plot(ax=axes[2], cmap="Purples", colorbar=False)
axes[2].set_title("Fusion Model")

plt.suptitle("Model Comparison — Confusion Matrices", fontsize=16)

plt.savefig("confusion_matrices.png", dpi=300, bbox_inches="tight")
plt.show()

print("Confusion matrices saved as confusion_matrices.png")
```




```
Confusion matrices saved as confusion_matrices.png
```

```
!cp performance_metrics.csv /content/drive/MyDrive/
!cp performance_chart.png /content/drive/MyDrive/
!cp confusion_matrices.png /content/drive/MyDrive/

print("Files copied to Google Drive")
```

```
Files copied to Google Drive
```

Start coding or generate with AI.